LESSON PLAN

GAS TURBINE ENGINE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	14
Total No. of Lesson Plans	16	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Lubrication system main components and its purposes, operation of each components of system, Air cooling and sealing of engine main bearings
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Lubrication system main components and its purposes, operation of each components of system, Air cooling and sealing of engine main bearings.
5. Specific Objectives	13.1.6.1 Trainee will be able to list the components of engine lubrication system and their function.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Lubrication System Schematic Diagram on Wall Chart; Instructor supervision required for component identification.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of engine lubrication?
(b) Name two types of lubrication systems used in aircraft engines.
(c) Why is oil circulation critical for bearing longevity?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Oil Reservoir and Scavenge Pumps • Pressure and Scavenge Filters • Oil Cooler and Heat Exchanger • Oil Lines and Check Valves • Lubrication of Main and Accessory Bearings	L & Ds	PPT, IB	15 min
(b)	• Operation of Pressure Lubrication Circuit • Function of Scavenge System Components • Air Cooling Mechanism for Oil • Air Sealing of Main Bearings • System Pressure Regulation and Monitoring	L & Ds	PPT, PDF	20 min
(c)	• Maintenance of Lubrication System Filters • Troubleshooting Low Oil Pressure • Inspection of Air Seals and Cooling Ducts • Interpretation of Oil Analysis Reports • Preventive Maintenance Procedures	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the air cooling system prevents oil overheating in high-performance engines.
(b) Describe the role of air seals in preventing bearing contamination and oil leakage.
(c) List three critical components of the scavenge system and their specific functions.

12. Home Assignment and Follow Up	Prepare a brief report comparing the lubrication requirements of a piston engine versus a gas turbine engine, focusing on oil flow rates and cooling methods, to be discussed in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Ice protection, hot air protection, electrical system	Refer to the master syllabus reference list	LP 15

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min